HAYAM ABDELBASET MAHMOUD REZK

AI/ML Researcher • Generative AI • LLMs • AI Agents • Responsible and Explainable AI

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Profile

Master's student in Data Science and Machine Learning at Queen's University, building practical AI/ML and GenAI systems through research and engineering projects, with a peer-reviewed paper on language-model reasoning over retrieved evidence. Works with Python, SQL, AWS, machine learning pipelines, LLM integration, and LoRA fine-tuning. Research interests include generative AI, large language models, AI agents, and responsible and explainable AI. Earlier work explored quantum computing, blockchain, and software engineering. Nearly a decade of university teaching includes 14+ computer science courses, a data encryption course with its own syllabus, and 15+ supervised student projects from research to working systems. Currently attending an AI Summit to meet researchers and explore collaborations.

Education

Master of Data Science and Machine Learning 2026–Present
Queen's University, School of Computing | Kingston, Canada

- Coursework: Deep Learning, Reinforcement Learning, Advanced Machine Learning, Big Data Systems, Natural Language Processing
- Research: evidence-grounded language models for legal text (supervised by Dr. Rahatara Ferdousi; industry mentor at Capgemini)

Master of Computer Science 2024–Present
Assiut University | Assiut, Egypt

- Research areas: Deep Learning, Natural Language Processing

B.Sc. Computer Science (Very Good) 2011–2015
Assiut University | Assiut, Egypt

Publications and Research

Evidence-Grounded Contract Clause Classification with Small Language Models NLLP 2026
Accepted, Natural Legal Language Processing Workshop (ACL Anthology) | with D. Mohamed and M. Anber

- Showed that retrieval-augmented small language models classify contract clauses on CUAD far more reliably than the same model alone: macro-F1 rose from 31.55% to 56.51% and accuracy from 46.50% to 70.35%, with no larger model and no retraining
- Built an offline pipeline (FAISS, Ollama, Qwen 2.5, Llama 3.2) designed for privacy-sensitive settings
- Extending the study to 1,679 clauses with fine-tuned encoder baselines (BERT, Legal-BERT), kNN variants, McNemar's tests, and bootstrap confidence intervals

Experience

Teaching Assistant 2016–2025
Egyptian E-Learning University | Sohag, Egypt

- Taught 14+ computer science courses (programming, data structures, OOP, algorithms, software engineering) to 500+ students
- Supervised 15+ capstone projects from research and design to working systems, spanning AI/ML, NLP, IoT, cybersecurity, and educational technology
- Designed laboratory materials aligned with current industry tools and development practice

Software Developer 2018–2019
3545 Consulting (U.S. company) | Remote

- Maintained and debugged C# and SQL applications, optimized database queries, and delivered client-server updates with senior developers

Selected Projects

ThinkGen: Adaptive AI Tutoring 2026
Python, Streamlit, Groq API, LLaMA-3.3-70B

- Built a tutoring application using Socratic questioning and dynamic difficulty adjustment, with progress tracking and personalized interaction

Biomedical Research Chatbot 2026
AWS EMR, Spark, Terraform, Docker, LoRA, HuggingFace
- Built a four-phase MLOps pipeline (ingestion, Spark preprocessing, LoRA fine-tuning, deployment); fine-tuned TinyLlama-1.1B with 4.5M trainable parameters and deployed it on AWS

TTC Subway Delay Prediction 2025
Python, Scikit-learn, Pandas
- Predicted delays from 25,000 incident records using 57 engineered features; compared 7 classifiers (best: LightGBM, F1 = 0.75) with chronological validation and class-imbalance handling

AWS Big Data and Load Balancing 2026
AWS EMR, PySpark, Terraform, ALB
- Processed the 25M-row MovieLens dataset with distributed Spark and built load-balanced infrastructure with EC2 failover

AI-Powered Text Summarizer 2015
Graduation project | Python, NLP
- Built a desktop application that extracts key points from English text

Selected Supervised Projects

Smart Book Framework 2025
Capstone supervision | Machine Learning, NLP
- Guided an AI-based learning platform that generates quizzes and surfaces relevant study material for the current topic

Recipe Recommendation System 2025
Capstone supervision | Machine Learning
- Oversaw an ML recommender that suggests recipes from available ingredients, with meal and dietary planning

Conversational AI Character 2024
Capstone supervision | Machine Learning, NLP
- Supervised a conversational assistant that pairs NLP with a digital avatar for interactive communication

Sentinel Home Security 2024
Capstone supervision | IoT, AI
- Mentored an AI-enhanced home monitoring system with real-time alerts

Technical Skills

- **Languages and tools:** Python, SQL, C#, Git, Linux, Jupyter
- **Machine learning:** Scikit-learn, Pandas, NumPy, model training and evaluation, statistical testing, feature engineering
- **Language models:** HuggingFace, LoRA fine-tuning, retrieval-augmented generation, FAISS, Ollama, Groq API
- **Cloud and systems:** AWS (EMR, S3, EC2), Terraform, Docker

Certifications

- Deep Learning Specialization
- Machine Learning Specialization
- AWS Certified Cloud Practitioner
- Python for Data Science, AI & Development
- Tools of the Trade: Linux and SQL
- Introduction to Data Engineering

Activities

1st Place, Faculty C++ Competition (Team Leader) 2013
Cloud Computing Ambassador, Assiut University 2014
Microsoft Student Partner and IEEE Member 2013–2015
Attendee, Arab Physics Society Conference 2023

Languages

Arabic Native
English Professional working proficiency